



School of Computer Sciences
CCA103 Systems Analysis and Design

PROJECT GUIDELINES

I. Overview

This document provides information and guidelines on how to prepare and implement the project for the subject of CCA103: Systems Analysis and Design.

II. Project Objectives

The project objectives are as follows:

- To demonstrate report writing and leadership skills for the system development project in a group.
- To inculcate concepts of systems requirements, analysis, design, and development process.
- To be able to present the analysis and design of the proposed system.

There are six core processes required in the development of any new software system:

- **C1.** Identify the problem and obtain approval
- **C2.** Plan and monitor the project
- **C3.** Discover and understand the details (Analysis Activities)
- **C4.** Design the system components that solve the problem
- **C5.** Build, test and integrate system components
- **C6.** Complete system tests and deploy the solution.

III. Project Groups Formation

You are to work in a group of **four (4)** members. No individual project is allowed.

All the members in a group must indicate the group formation on the spreadsheet given by the lecturers. Once the group is formed, one of the group members will be selected as the group leader.

Update the given spreadsheet indicating **group number, full name of all group members and matric number, project leader, project title and SDG chosen.**

IV. Selection of Project Title

You are required to design and develop a system (software system). **Please ensure that the title and your project can be mapped to one of the SDGs.**

Every group must propose one title. Once your group has decided on your project title, please update your group post on the spreadsheet to include the project title and a short description of the project. Be as specific as possible when you are posting your project title. For example, the project title “University Book Library System” instead of merely “Library System” will give us a better idea of what your group is planning to work on. Everyone in the class should be able to view all the project titles posted on the spreadsheet. No title should be the same, meaning no more than one group can work on the exact same title.

Please check the spreadsheet before you post your group title to make sure that your project does not coincide exactly with another group that has already posted on the spreadsheet.

The approval of the project title will be based on a first-come-first-serve basis and will be posted on the comments section by the instructors on the spreadsheet.

Based on the selected title, each member of a group must be responsible for at least one module in the project. For each module, the following requirements must be fulfilled:

- Each member must demonstrate and develop their own use case within their modules.
(FINAL USE CASE must combine all these use cases from different modules)
- Each module must **have at least THREE use-cases. Some use cases may be overlapping with other modules but** ensure at least TWO are unique to only to the module.

All the modules must be integrated as a complete system. Each group has to demonstrate an overall system that consists of at least four modules at the end of this semester.

V. Deliverables and Important Dates

The project deliverables and important dates are as follows (see Table 1):

Deliverables	WEEK	Due dates	Report Submission
Group formation and project title	2	27 March 2026, Friday	Shared Spreadsheet
Project proposal & Scope	3	3 April 2026, Friday	PDF Softcopy
Report 1: System Analysis	9	9 May 2026, Saturday	PDF Softcopy
Report 2: System Design & Implementation	15	23 June 2026, Tuesday	PDF Softcopy
System demo/presentation	15-16	24 -30 June 2026	PDF Slides & Demo on Webex

Table 1: Project deliverables and important dates.

Submission procedure: All the reports must be submitted by the mentioned due dates listed in Table 1. Submit a soft copy of your report in the assignment on eLearn@USM. **Late reports** without prior approval **will be penalised**. Every group needs to conduct a system demo at the

end of the semester. Each member must participate in the system demo and explain the module(s) done by him/her. Details on the system demo (e.g. procedures, time, venue, etc.) will be made available towards the end of the semester.

VI. Documentation Format and Guidelines

This section explains the documentation format and guidelines on how to prepare the documentation for the proposal, system analysis and system design.

Report Format:

All reports must comply with the following format:

- Font: Times New Roman.
- Font size: 11.
- 1.5 lines spacing.
- o Page size: A4 (8.27" x 11.69").
- All pages must be properly numbered.
- For all reports (Report 1: System analysis and Report 2: System design), a cover page must be included. A report with no cover page will not be marked. Refer to the sample cover page.
- All figures and tables (if any) should be labelled so that cross-referencing can be done in text.
- A table of contents (TOC) must be included.
- **Proper references** and bibliography must be included.
- Some parts have many diagrams to be included. Include example (2-3 diagrams), and for the rest, put the references to which appendices (number the appendix appropriately), and place the diagrams in the appendices. Be systematic in numbering and referring to the appendix.
- **PLEASE FOLLOW THE FORMAT** and give attention to the format, as marks will be given for proper formatting.



SCHOOL OF COMPUTER SCIENCES
UNIVERSITI SAINS MALAYSIA

CCA103 : Systems Analysis and Design
Semester II, Academic Session: 2025/2026

Project Proposal / System Analysis / System Design

[Group Number]

[Project Title]

Names	Matric No	USM Email Addressess	Module
Rasheed Sidque (Leader)	10001	rashss@student.usm.my	Ac...
Lim Chung Wey	10002	lcw99@student.usm.my	DEF
Thiinaah Murali	10003	thiinaah@student.usm.my	GHI
Pandaleela Rina	10004	pLeela@student.usm.my	JKL

Date of submission
[Date - Month 2026]

Submission #1

Project Proposal & Scope:

It is compulsory to submit this document for the project title and scope to be approved.

The project proposal & scope should contain the following:

1. A project title.
2. A list of team members (name, matric number, USM email, role, major/minor)
3. An executive summary (overall project summary)
 - a. A brief description of the organisation you are developing the system for
 - b. Project scope.
4. Please highlight the **SDG** related to the project.
5. List of modules which you intend to implement as part of the proposed project.
 - a. For each of these modules, a brief description is expected.
 - b. Indicate the module(s) assigned to each group member.

Submission #1 □ 15 % of the Coursework

REPORT 1

System Analysis:

The linkage between sections should be stated clearly at the beginning and/or the end of each section to show the relationships between sections. The report should contain the following:

- a. Cover page.
- b. Table of Contents (TOC)

c. Abstract (2%)

This should be no more than one page in length. The abstract should allow the reader who is unfamiliar with the work to gain a swift and accurate impression of what the project is about, how it arose, and what has been achieved.

An abstract is a project summary that gives a basic description of the system, its objectives, and target users. It gives a brief overview of the problems to be solved, how to solve them, and the products of the project/solution. Abstract is not an introduction to the chapters or an explanation of the course.

d. Chapter 1: Introduction (15%)

- Organization/business background & problem description
- System objectives
- Proposed solution
 - With the breakout of modules
 - Proposed system description and capabilities (in modules)
- Benefits and Uniqueness of the Proposed Solutions
- System scope and limitation
- SDG ALIGNMENT

e. Chapter 2: Project Planning and Project Management (30%)

- Development methodology
- The organizational, technological, resource, and schedule risks and feasibility
- Project management plan
 - Hierarchy chart of work division among group members
 - Project iteration schedule (in table form)
 - SWOT Analysis
 - Work breakdown structure (WBS) – PMBOK style
 - Gantt chart

f. Chapter 3: System Requirements, Analysis and Modelling (45%)

- Existing systems review - problems and weaknesses in the current systems.
- Summary of any relevant systems to provide context, key terms, and concept so readers can understand the proposed system.
- Information gathering techniques – Provide the stakeholder list and describe how you get your requirements from stakeholders
- System requirements - provide detailed requirements of the new system. List down the system functionalities and features
- UML diagrams (with some explanation):
 - * Use-case diagram (for each module)
 - * Use-case descriptions - for each use case

- * Activity diagrams for complex use-cases (at least 1-2 from each module)
- * Domain class diagram
- * System sequence diagram for each use case
- * State Chart Diagram (1-2 overall)

g. Chapter 4: System Design Strategy (5%)

- High-level design - an overview of the network/proposed system architecture

h. References

Formatting, TOC & References (3%)

Submission #2 □ 10 % of the Coursework

REPORT 2

System Design & Implementation:

- Cover page.
- Table of Contents (TOC)
- a. **Abstract (2%)** – Update and revise abstract
- b. Chapter 1 (From Report 1)
- c. Chapter 2 (From Report 1)
- d. Chapter 3 (From Report 1)
 - **NEW CONTENT** □ Overall USE CASE Diagram for the system **(5%)**
- e. **Chapter 4: System Design (55%)**
 - Architectural Design (High-level design)
 - o Enterprise-level systems, Client-Server systems or Internet-based systems
 - Hardware and software specifications
 - Detailed Design using UML
 - o Detailed Sequence diagrams (view, domain, and data access layers)
 - o Design class diagram (with methods and controller classes)
 - o Package Diagram
 - User Interface Design
 - o Overall Menu Hierarchy
 - o Windows/mobile/browser screen design
 - o Forms (input) and reports (output) design
 - Design system interface
 - o Integrity Control
 - o Security Control
- f. **Chapter 5: System Implementation and Testing (20%)**
 - Development order (Input, process, output (IPO), top down or bottom up)
 - Testing strategy (unit, integration, system, and acceptance tests)
- g. **Chapter 6: Conclusion (10%)**
 - Summary of how requirements have been met
 - Outline the success of the project when compared to the objectives that were set
 - Provide a critical evaluation of the project (advantages and disadvantages) by comparing it with existing systems.
 - Suggest future work for the project
- h. **References**
- *Formatting, TOC & References (3%)*
- *Overall flow from System Analysis (report 1) to System Design and Implementation (Report 2) (5%)*

Submission #3 □ 5% of the Coursework

Presentation

- I. Slides should be submitted in PDF format
Online Presentation:
- II. Please stand by 5 minutes before the presentation, and make sure all members are present and ready. Each member must present some parts of the presentation
- III. Please prepare the slides, and the presentation should be done in 10-12 minutes.
- IV. The contents of your presentation should cover at least:
 - The introduction with objectives
 - The modules – who oversees each module
 - Demo of system analysis design

Presentation/Demo evaluation. Please refer to the full Rubrics.

1. Overall Content of the slides (5%)

2. System analysis design (75%):

- a. *Project Management (15%)*
- b. *System Use Cases (Overall) & Detailed Sequence Diagram (1 each module) (20%)*
- c. *Activity Diagram & State chart Diagram (1 each module) (10%)*
- d. *User Interface (prototype) (2-3 screenshots for each module) (20%)*
- e. *System completeness – implementation strategy (10%)*

3. Clear and confident delivery of ideas (10%)

4. Understand and respond to questions (10%)

VII. Project Evaluation Rubrics

The project contributes 30% of your overall grade. The marks distribution of the project is as follows:

Assignment (Project)	30%
• Group Formation & Proposal • Report 1 • Report 2 • Presentation/ Demo	• 15% • 10% • 5%

The system analysis, system design & implementation, and system demonstration will be evaluated based on their respective grading rubrics. The project report will be evaluated as **a group submission**.

Students are expected to divide the project modules fairly among group members. If the module distribution is not balanced, it may affect the overall group marks, regardless of individual effort.

In situations where one member's contribution is significantly lower than the rest of the group, the evaluation may also consider the individual's selected module. However, since this is a group project, some marks will still affect the entire group, especially if the workload was not properly managed among members.

If there is a group member who is irresponsible or unresponsive (for example, cannot be contacted or does not participate), **the group must inform the lecturers** before the report submission. This is important to prevent the entire group from being negatively affected.

RUBRICS

Report 1

Criteria	Weight	A, A – 10% (9-10) Superior Command	B+, B, B- 10% (6-8) Good Command	C+, C, C- 10% (3-5) Fair/Some Command	D+, D, D-, F 10% (0-2) Poor or No Command
Abstract	2%	The abstract is clearly stated, has a sharp and distinct focus	The abstract section is clearly stated with a focus	The abstract section is quite vague	The abstract section is vague and very weak.
Introduction	15%	Presented a very good introduction of the domain with key aspects highlighted.	Presented a good introduction of the domain. Some key aspects were highlighted	Presented a fair introduction of the domain. Key aspects were not well highlighted.	Presented a poor introduction of the domain. Key aspects were not highlighted.
Project Planning and Project Management	30%	Presented a very good coverage of project planning and management. Details were well presented.	Presented a good coverage of project planning and management. Some details were presented.	Presented a fair coverage of project planning and management. Details were not well presented.	Presented a poor coverage of project planning and management. Details were not presented.
System Requirements, Analysis and Modelling	45%	Analysis was sufficient and very well presented.	Analysis was sufficient and well presented.	Analysis was fair and well presented.	Analysis was poor and not well presented.
Design Strategy	5%	Presented a very good system design. Details were presented.	Presented a good system design. Some details were presented.	Presented a fair system design. Details were not well presented.	Presented a poor system design. Details were not presented.
Organization (Formatting & References)	3%	Report is well organized, easy to read, and very neat.	Report shows adequate organization, readable, and neat.	Report is poorly organized, difficult to read, and somewhat messy.	Report has no organization, not easy to read, and messy.

Report 2

Criteria	Weight	A, A – 10% (9-10) Superior Command	B+, B, B- 10% (6-8) Good Command	C+, C, C- 10% (3-5) Fair/Some Command	D+, D, D-, F 10% (0-2) Poor or No Command
Abstract	2%	Summarizes the goal of the project and results concisely. Specific information provided.	Most information is presented, but some items may be missing.	Insufficient summary of what was done. Few specific results were presented.	Abstract is missing or contains no specific information.
System Analysis (Use Case only)	5%	The use case diagram was sufficient and very well presented.	The use case diagram was sufficient and well presented.	The use case diagram was fair and well presented.	The use case diagram was poor and not well presented.
System Design	55%	Presented a very good system design. Details were presented.	Presented a good system design. Some details were presented.	Presented a fair system design. Details were not well presented.	Presented a poor system design. Details were not presented.
System Implementation and Testing	20%	Presented a very good system implementation and testing. Details were presented.	Presented a good system implementation and testing. Some details were presented.	Presented a fair system implementation and testing. Details were not well presented.	Presented a poor system implementation and testing. Details were not presented.
Conclusion	10%	The conclusion was sufficient and very well presented.	The conclusion was sufficient and well presented.	The conclusion was brief and well presented.	The conclusion was too brief and not well presented.
Overall Flow	5%	The overall report from Chapter 1 to the conclusion was very well presented.	The overall report from Chapter 1 to the conclusion was well presented.	The overall report from Chapter 1 to the conclusion was not well presented.	The overall report from Chapter 1 to the conclusion was very weak.
Organization & Formatting (Neatness)	3%	The report is well organized, easy to read, and very neat.	The report shows adequate organization, readability, and neatness.	The report is poorly organized, difficult to read, and somewhat messy.	The report has no organization, is not easy to read, and is messy.

RUBRICS for Presentation

Criteria	%	Very good	Good	Fair	Weak	Very weak
		9-10	7-8	5-6	3-4	1-2
Overall Content of the slides (5%)	5	Slides with relevant information. The topic is discussed clearly, and adequate points are made.	Slides with relevant information. The topic is somewhat discussed but more information is required.	Slides with relevant information. The topic is slightly discussed but more material is needed.	Less slides that contain relevant information. The topic is addressed but not discussed or elaborated on.	Slides do not contain relevant information. The topic is not addressed, discussed, or elaborated on.
System design& implementation (75%):						
<i>Project Management (15%)</i>	15	The student has created an EXCELLENT management plan.	The student has created a GOOD management plan.	The student has created an ACCEPTABLE management plan	The student has created a WEAK management plan	The student has created an UNACCEPTABLE management plan.
<i>System Use Cases & Detailed Sequence Diagram (20%)</i>	20	The student has created an EXCELLENT use case and sequence diagrams.	The student has created a GOOD use case and sequence diagrams.	The student has created an ACCEPTABLE use case and sequence diagrams.	The student has created a weak use case and sequence diagrams.	The student has created an UNACCEPTABLE use case and sequence diagrams.
<i>Activity Diagram & State chart Diagram (10%)</i>	10	The student has created EXCELLENT diagrams.	The student has created GOOD diagrams.	The student has created acceptable diagrams.	The student has created weak diagrams.	The student has created UNACCEPTABLE diagrams.
<i>User Interface (prototype) (20%)</i>	25	The student has created an EXCELLENT user interface, which includes ALL or above 90% of the mentioned features.	The student has created a GOOD user interface, which includes 70 - 90% of the mentioned features.	The student has created an ACCEPTABLE user interface, which includes some of the mentioned features (50%-70%).	The student has created a WEAK user interface, which includes some of the mentioned features (30%-50%).	The student has created an UNACCEPTABLE user interface, which includes some of the mentioned features (less than30%).

System completeness (10%)	10	The system is fully complete with an excellent strategy.	The system is partially complete with a good strategy.	The system is partially complete with an acceptable strategy.	The system is incomplete and also has a weak strategy.	The system is incomplete with an unacceptable strategy
Clear and confident delivery of ideas (10%)	10	Able to deliver ideas with great clarity and confidence	Able to deliver ideas clearly and confidently	Able to deliver ideas fairly clearly and confidently, thus require minor improvements	Able to deliver ideas with limited confidence and require further improvements	Not able to deliver ideas clearly and confidently and require major improvements
Understand and respond to questions (10%)	10	Able to fully understand and respond to questions very well	Able to respond to questions well	Able to understand and answer questions satisfactorily	Able to understand and answer questions but not able to accurately answer the question	Not able to understand and respond to a question